

78. human STXBP5 REPEAT "WD 5" (33 aa) and zebrafish Q5SQE2

TOOLS: kmerseek hp_kyte_doolittle2 k19 (chosen for REPEAT in Stage 0) and hp_thomas_dill2 k19; phmmer, MMseqs2, MMseqs2 iterative, Foldseek, ProstT5, Reseek; Kyte-Doolittle scan

- kmerseek: call lands on the feature ($\geq 80\%$ of call inside, $\geq 30\%$ covered)

call mostly inside the feature, lower IoU than kmerseek

call overlaps the feature but $< 50\%$ of the call is inside it

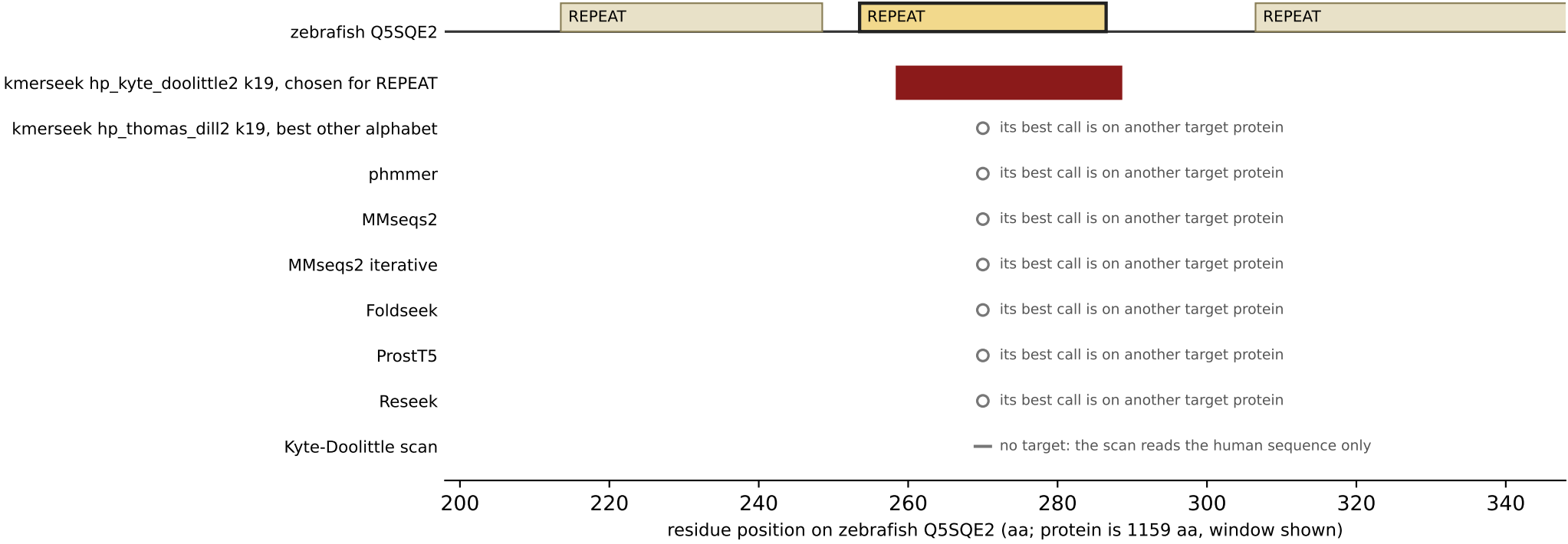
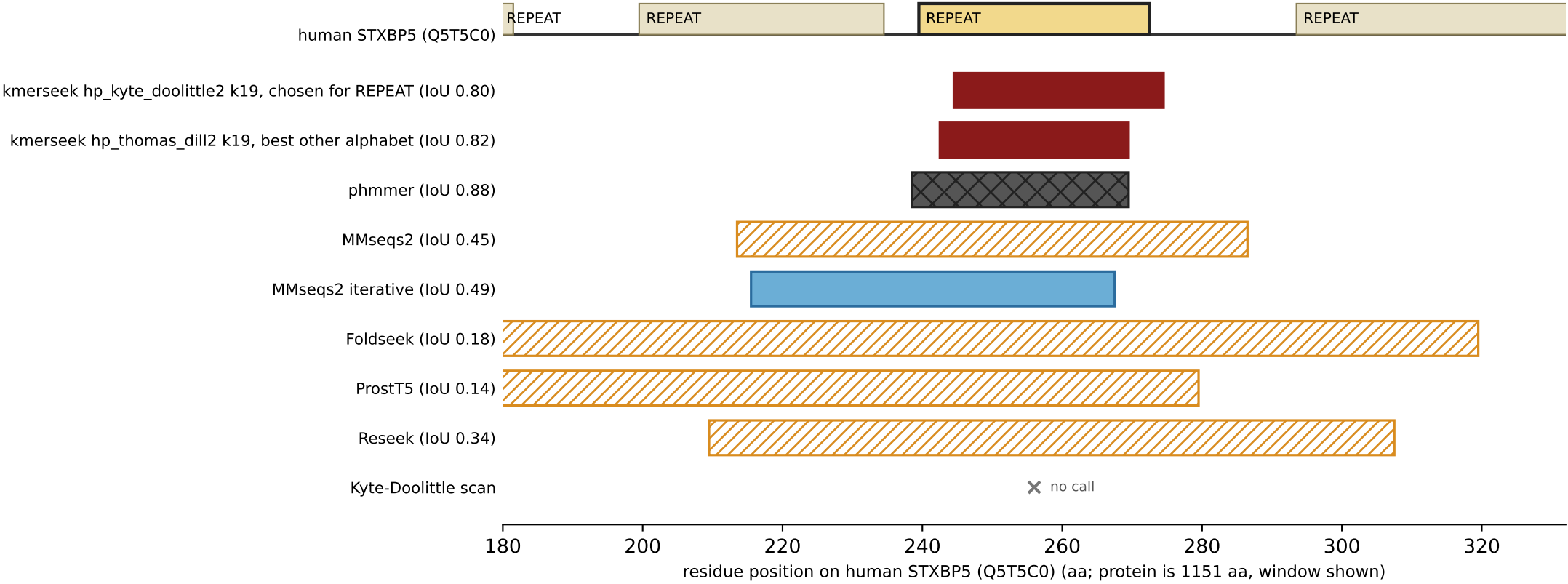
call reaches kmerseek's IoU or higher

no call of this feature type overlaps the feature
- the tool's best call is on another target protein

Kyte-Doolittle scan: no target protein, it searches nothing

the Swiss-Prot feature being scored (human) / transferred from (target)

other Swiss-Prot features, labelled by type



Hypothesis: kmerseek places the human REPEAT feature "WD 5" on its zebrafish counterpart and the structure tools do not.
Conclusion: kmerseek IoU 0.80, 21 of 30 residues identical; Foldseek spills; ProstT5 spills; Reseek spills; phmmer equal or higher IoU; MMseqs2 spills; MMseqs2 iterative inside, lower IoU; Kyte-Doolittle scan did not land; pLDDT 97, disorder fraction 0.0.